No-Code Platform User Manual

1. Source Data

1. Launching the Section

From the dashboard sidebar, click **Source Data**.

2. Choose the Data Input Method

In the upload panel, choose one of the available input types:

- **Local**
- **URL**
- **Database connection**

This determines how the app will read and register your dataset.

3. Enter Study Information

Before uploading, complete the study metadata fields:

- **Study Name**
- **Study Country**
- **Additional Info**

These details are saved with the dataset and help you identify it later in the uploaded data logs.

4. Upload a File or Connect to a Database

If you selected **Local**, click **Browse** and choose your dataset file.

If you selected **URL**, paste the dataset link into the upload field.

If you selected **Database connection**, provide the connection details:

- database type
- host
- database name
- username
- password
- port

Then click **Connect**.

5. Choose a Table or Run a Query

After connecting to a database, choose whether to:

- select a schema/table directly, or
- run a custom query

The app will display a preview of the returned data so you can confirm that the correct table or query result has been loaded.

6. Submit the Upload

Click **Submit Upload**.

The app will:

- validate the data source
- read the dataset
- generate metadata
- save the uploaded file or table to your workspace
- register the upload in the logs

7. Review Uploaded Datasets

Use **Show Uploaded** to display the uploaded datasets table.

This table helps you confirm:

- what has already been uploaded
- the metadata attached to each dataset
- which datasets are available for later sections

8. Use Undo / Redo

The Source Data section includes history controls.

Use these to move backward or forward through recent changes to:

- upload type
- metadata entries
- database connection fields
- selected schema/table/query settings

Tips & Best Practices

1. Always confirm the dataset appears in the uploaded logs before moving to Manage Data.
2. When using a database connection, preview the returned data before clicking Submit Upload.
3. Use meaningful study names so datasets are easy to identify later.

2. Manage Data

Manage Data is divided into four sub-sections:

- **Overview**
- **Explore**
- **Transform**
- **Combine Data**

2.1 Overview

1. Open the Overview Section

From the sidebar, go to **Manage Data** then **Overview**.

This is the starting point for selecting the active dataset.

2. Select the Dataset

Choose the dataset from the dataset selector.

Then click **Apply**.

The selected dataset becomes the active working dataset for the rest of the dashboard.

3. Choose the Summary Display

Use the **Show** option to select how the dataset summary should be displayed.

The app supports several summary formats, including:

- structure-style output
- base summary output
- skim-style summary
- summarytools-style summary

4. Review the Dataset Structure

The summary panel displays information such as:

- variable names
- data types
- missing values
- basic descriptive statistics
- overall dataset structure

5. Change the Active Dataset When Needed

To work on a different uploaded dataset, select a new dataset and click **Apply** again.

The working environment updates to the newly selected dataset.

Tips & Best Practices

1. Always begin here before using Explore, Transform, Visualize Data, Research Questions, or Machine Learning.

2.2 Explore

1. Open the Explore Section

Go to **Manage Data** then **Explore**.

This section helps you inspect the active working dataset in more detail.

2. Show the Dataset

Enable **Show Data**.

Then choose how the data should be displayed.

If you select **All**, the app shows a full interactive table.
If you choose another display type, the app shows a printed/text version of the dataset output.

3. Inspect Missingness

Enable **Missingness** to review missing-data information.

This allows you to identify variables that may need cleaning before transformation or modelling.

4. Filter the Dataset

Enable **Filter**.

A filter rules input area appears where you can enter filtering conditions.

Click **Apply Filter** to subset the working dataset.

Use **Reset Filter** to remove the applied filter and restore the full working dataset.

5. Select Variables

Enable **Select Variables**.

Choose the variables you want to focus on for exploration.

This is useful when you want to narrow the working dataset to only the variables needed for a specific analysis.

6. Run Quick Explore

Enable **Quick Explore**.

The app generates a quick descriptive summary of the current working dataset, including any filters or selected variables already applied.

7. Update the Working Dataset

Use the update option if you want the explored version of the data to become the current working copy.

This is particularly useful after filtering or variable selection.

8. Use Undo / Redo

The Explore section includes history controls so you can step backward or forward through recent exploration changes.

Tips & Best Practices

1. Apply filters before running Quick Explore for cleaner summaries.
2. Use variable selection to simplify later visualization and modelling steps.
3. Reset filters if the resulting dataset becomes smaller than expected.

2.3 Transform

1. Open the Transform Section

Go to **Manage Data** then **Transform**.

Choose the variable you want to work on from the variable selector.

Most transformation tools become active only after a variable is selected.

2. Change Variable Type

Enable **Change Type**.

Choose the new type for the selected variable, then apply the change.

This is useful when a variable has been imported with the wrong type, for example text that should be numeric or a category that should be a factor.

3. Rename a Variable

Enable **Rename Variable**.

Enter the new variable name and apply the change.

The working dataset is updated and the rename is recorded in the transformation log.

4. Recode Variable Values

Enable **Recode Variable**.

Choose the value or label to change, enter the replacement value, and apply the recode.

This is useful for cleaning category labels and harmonizing values before plotting or modelling.

5. Create Missing Values

Enable **Create Missing Values**.

Then choose whether to create missing values:

- for the selected variable, or
- across all variables

Next choose how values should be identified:

- by **pattern**
- by **category**

Depending on the selection, you can enter:

- text patterns
- category labels

- numeric values
- range conditions

Apply the action to convert the chosen values into missing values.

6. Identify and Handle Outliers

Enable **Identify Outliers**.

For numeric variables, the app detects outliers and displays supporting outputs.

Then choose how to handle them, for example:

- keep them unchanged
- drop them
- replace or correct them using a selected value such as mean or median
- enter a custom corrective value where applicable

Apply the selected method to update the dataset.

7. Handle Existing Missing Values

Enable **Handle Missing Values**.

Choose whether to manage missing values:

- for the selected variable, or
- across all variables

The app supports actions such as:

- dropping missing records
- creating a new category label
- assigning a new numeric replacement value

Apply the action to update the working dataset.

8. Review Logs and Diagnostics

The Transform section displays several supporting outputs, including:

- variable type information
- rename logs
- recode logs
- missing-value creation logs
- outlier-handling logs

- missing-value handling outputs
- quick summaries
- quick plots

9. Use Undo / Redo

Use the Transform history controls to move backward or forward through your recent changes.

Tips & Best Practices

1. Correct variable types before running plots or models.
2. Recode inconsistent category labels before analysis.
3. Review the logs after each step so you can confirm exactly what changed.

2.4 Combine Data

1. Open the Combine Data Section

Go to **Manage Data** then **Combine Data**.

Use this section to merge or append another uploaded dataset to the current working dataset.

2. Select the Dataset to Add

Choose the second dataset from the dataset list.

The current working dataset acts as the base dataset.

3. Choose the Combination Type

Select how the datasets should be combined.

This may involve combining rows or combining columns, depending on your selected merge approach.

4. Choose the Match Type

Choose how variables will be matched between the two datasets.

You can use:

- automatic matching
- manual matching

5. Review Base, New, and Matched Variables

The app displays three variable lists:

- variables in the base dataset
- variables in the incoming dataset
- matched variables

This helps you see which fields are already aligned.

6. Match Variables Manually When Needed

If variables have the same meaning but different names, use the manual matching controls.

Select the base variable and the corresponding variable in the new dataset, then apply the match.

The app aligns the variable names for merging.

7. Create an ID Variable if Required

If a merge requires a record identifier, enter or generate the ID variable using the provided input.

This is especially useful for record-level merging.

8. Perform the Merge

Click the merge/apply button to complete the combination.

The app updates the working dataset and records the operation in the merge log output.

9. Use Undo / Redo

Use the Combine Data history controls to step backward or forward through your merge setup changes.

Tips & Best Practices

1. Check matched variables carefully before applying the final merge.
2. Use manual matching whenever the same concept is stored under different variable names.
3. Ensure an ID field is available when combining by records rather than stacking rows.

3. Visualize Data

Visualize Data has two sub-sections:

- **Automatic**
- **Custom**

3.1 Automatic

1. Open the Automatic Visualization Section

Go to **Visualize Data** then **Automatic**.

This section is designed for fast, ready-made visual summaries.

2. Create a Bivariate Plot

In the Bivariate panel:

- choose the **outcome** variable
- choose one or more **feature** variables
- choose the **color palette**
- optionally enter the plot title
- click the generate button

The app creates a large bivariate plot in the output panel.

3. Create a Correlation Plot

In the Correlation Plot panel:

- choose the variables to include
- choose the color palette

The app generates a correlation plot showing the relationships among the selected features.

4. Download the Automatic Report

Use the download button to export the automatic visualization report.

Tips & Best Practices

1. Use bivariate plots after defining your target variable in the modelling setup.
2. Use the correlation plot before model training to detect highly related predictors.

3.2 Custom

1. Open the Custom Visualization Section

Go to **Visualize Data** then **Custom**.

This section allows you to build more detailed tables and graphs.

2. Choose the Output Type

Select whether you want to create:

- a **table**
- a **graph**

The controls shown on the page will update based on your choice.

3. Create a Table

If you choose a table, select the relevant options such as:

- calculation variable
- row variable
- table option/tab type

Then click **Create Cross Tab**.

You can refine the output using options such as:

- p-values
- confidence intervals
- numeric summary style
- dropping missing values
- caption/title settings

The generated table appears in the output panel.

4. Create a Graph

If you choose a graph:

- choose the plot type
- choose the x variable
- choose the y variable where required
- enter the title and axis labels
- click **Create**

The plot is displayed in the main output area.

5. Customize the Graph

The custom graph tools allow you to adjust many formatting options, including:

- color variable
- grouping variable
- theme
- bar width
- line size and line type
- point shape
- smoothing
- confidence intervals
- summary method
- title size and position
- axis text size
- facet title size
- x-axis text angle
- legend title
- stacking
- density overlays
- palette selection

6. Review the Data Preview

The section also includes a data preview table so you can verify that the chosen variables and values are correct before finalizing the chart or table.

7. Download the Output

Use the download controls to save your custom table or graph.

Tips & Best Practices

1. Use Automatic for quick summaries and Custom for publication-ready outputs.
2. Start with simple plots, then add formatting options gradually.
3. Always check the data preview before exporting final visuals.

4. OMOP

This section includes:

- **Evidence Quality**
- **Achilles Analysis**
- **OMOP Visualization**

- **Cohort Constructor**
- **Feature Extraction**

4.1 Evidence Quality

1. Open the Evidence Quality Section

Go to **OMOP** then **Evidence Quality**.

This section is used for OMOP data quality assessment.

2. Confirm the Database Connection

The OMOP tools rely on an active database connection, usually established in the Source Data section.

Make sure the database is connected before proceeding.

3. Select the OMOP Schemas

Provide the required OMOP schema details, including:

- CDM schema
- results schema
- source name

These are used to run the OMOP data quality checks.

4. Generate the Data Quality Dashboard

Click **Generate DQD**.

The app runs the OMOP Data Quality Dashboard process and writes the results to the user output directory.

5. Review the Process Logs

If there are schema or execution issues, the section displays the corresponding logs or error messages.

6. View the DQD Output

Click **View DQD** to open the most recently generated quality output.

Tips & Best Practices

1. Run Evidence Quality before cohort building or OMOP feature extraction.
2. Make sure the CDM schema and results schema are correct before running the checks.

4.2 Achilles Analysis

1. Open the Achilles Section

Go to **OMOP** then **Achilles Analysis**.

This section runs OMOP descriptive characterization using Achilles.

2. Select the Required Schema Inputs

Choose the schema details needed for the Achilles run.

3. Run Achilles

Click **Run Achilles**.

The app executes the Achilles workflow and generates descriptive OMOP outputs.

4. Review the Output

Use the generated summaries and outputs to inspect the structure and characteristics of the OMOP database.

Tips & Best Practices

1. Run Achilles after verifying the OMOP connection and schema details.
2. Use it alongside Evidence Quality for a more complete view of OMOP data readiness.

4.3 OMOP Visualization

1. Open the OMOP Visualization Section

Go to **OMOP** then **OMOP Visualization**.

This section provides interactive summaries of OMOP tables and domains.

2. Select Schemas and CDM Version

Choose the required inputs, including:

- CDM schema
- results schema
- vocabulary schema where needed
- CDM version

3. Generate Summaries

Click **Generate Summaries**.

The dashboard then prepares OMOP summary outputs.

4. Review the Main Summary Tabs

The app provides high-level summary views such as:

- **General Info**
- **Record Counts**
- **Unmapped Concepts**
- **Domain Distribution**

These give you a quick picture of the OMOP structure and content.

5. Select a CDM Table for Table-specific Analysis

Choose a table from the CDM table selector.

The app then displays table-specific analyses for the selected OMOP table.

6. Review Table-specific Outputs

Depending on the table selected, outputs may include:

- summary tables
- age distributions
- sex/gender distributions
- race or ethnicity distributions
- visit summaries over time
- concept frequency plots
- other table-specific visual summaries

Tips & Best Practices

1. Start with the main summary tabs before drilling into individual CDM tables.

2. Use the table-specific panels to focus on the domains most relevant to your study.

4.4 Cohort Constructor

1. Open the Cohort Constructor Section

Go to **OMOP** then **Cohort Constructor**.

This section is used to define and generate OMOP cohorts.

2. Create the CDM Reference

Under the CDM reference area, provide the necessary schema/connection details and create the CDM reference.

This initializes the OMOP context needed for cohort generation.

3. Define the Cohort Inputs

Once the reference is created, enter:

- the concept keyword
- the cohort name
- the cohort date

These values determine the cohort you want to generate.

4. Generate the Cohort

Click **Generate Cohort**.

The app processes the request and creates the cohort output.

5. Review the Cohort Summary

A cohort summary table is displayed once the process completes.

A CSV download option becomes available for the summary table.

6. Review the Cohort Visualizations

The section also generates cohort plots such as:

- gender plot
- age-group plot

- race plot
- ethnicity plot

A ZIP download option is provided when the plots are available.

Tips & Best Practices

1. Use clear cohort names and dates so outputs remain easy to track.
2. Confirm the OMOP vocabulary/concept information is available before cohort generation.

4.5 Feature Extraction

1. Open the Feature Extraction Section

Go to **OMOP** then **Feature Extraction**.

This section extracts OMOP covariates and saves them as a dataset for later modelling.

2. Confirm the Database Connection

If the database is not connected, return to Source Data and establish the OMOP database connection first.

3. Choose the OMOP Inputs

Select the required inputs:

- CDM schema
- results schema
- cohort table

Then choose the feature domains you want to extract.

These may include domains such as:

- demographics
- condition occurrence
- drug exposure
- measurement
- procedure occurrence
- observation

4. Review the Domain Record Summary

The app generates a record summary by domain for the selected cohort.

This helps you understand what data is available before running the extraction.

5. Name the Output File

Enter the output CSV name or file label.

6. Extract Features

Click **Extract Features**.

The app:

- configures OMOP covariate extraction
- extracts the selected features
- reshapes them into a person-level dataset
- saves the result
- registers it as a new uploaded dataset for later use

7. Download the Summary

Use the CSV download button to save the domain summary if needed.

Tips & Best Practices

1. Review the domain summary before extraction so you know which domains are populated.
2. Start with a smaller set of domains, then expand once the workflow is confirmed.

5. Research Questions

1. Open the Research Questions Section

Go to **Research Questions**.

This section uses AI-assisted prompting to generate research questions and additional analytical ideas from the active dataset.

2. Indicate Whether You Have an Outcome Variable

Choose whether your study includes an outcome variable.

If yes, select the outcome variable.

This helps tailor the generated research questions to a specific endpoint.

3. Enable Research Question Generation

Turn on the research question generation option.

The app will check whether a Gemini API key has been provided.

4. Enter the Gemini API Key

If no key is stored for the session, enter the Gemini API key and save it.

5. Generate Research Questions

Click **Generate Research Questions**.

The app uses the working dataset summary and quick-explore information to build a prompt and send it to Gemini.

The suggested research questions are displayed as text/markdown in the output area.

6. Generate Additional Suggested Analyses

Enable the **Additional Analysis** option if you want the app to suggest next analytical steps based on the previous AI-generated output.

7. Review and Use the Output

The section displays:

- candidate research questions
- additional suggested analyses

These outputs can guide your exploratory analysis, visualization plan, or modelling choices.

Tips & Best Practices

1. Run Explore or Quick Explore before using this section so the AI receives better dataset context.
2. Specify an outcome variable when you want more focused, hypothesis-driven suggestions.

6. Machine Learning

Machine Learning is organized into:

- **Initialize**

- **Feature Engineering**
- **Train Model**
- **Validate and Deploy Model**
- **Predict / Classify**

6.1 Initialize

1. Open the Initialize Section

Go to **Machine Learning** then **Initialize**.

This is where you define the modelling session.

2. Enter a Session Name

Provide a **session name**.

This identifies the modelling run and activates the session setup workflow.

3. Set the Seed

Enter the random seed value.

This supports reproducibility of the train/test split and modelling process.

4. Specify the Target Variable

Choose whether the dataset has a target variable.

If yes:

- select the target variable
- optionally choose variables to exclude from modelling

5. Choose the Analysis Type

Select the type of analysis you want to perform.

Depending on the setup, supported workflows may include:

- supervised learning
- clustering
- anomaly detection
- semi-supervised learning

6. Select Model Types

Choose the model family or specific model types you want to consider later during training.

7. Set the Partition Ratio

Use the partition slider to define the train/test split.

8. Apply the Setup

Click **Apply**.

The app stores the modelling setup for the Feature Engineering and Train Model stages.

Tips & Best Practices

1. Choose the target carefully because it determines the task type.
2. Exclude identifiers and leakage-prone variables before modelling.

6.2 Feature Engineering

1. Open the Feature Engineering Section

Go to **Machine Learning** then **Feature Engineering**.

2. Choose the Modelling Framework

Select the framework you want to use:

- **Caret**
- **PyCaret**

The available controls will change depending on your selection.

3. Configure Caret Feature Engineering

If using Caret, select the required preprocessing options, such as:

- partition strategy
- preprocessing on/off
- missing-data imputation
- imputation method
- feature-engineering options
- removal of highly correlated variables

- correlation threshold
- PCA
- upsampling
- upsampling method

4. Apply Feature Engineering

Click **Apply**.

The app prepares the processed train/test datasets and stores the transformed outputs for model training.

5. Review Preprocessing Logs and Plots

The page displays logs and supporting outputs, including class-balance or upsampling visualizations where relevant.

6. Configure PyCaret if Selected

If using PyCaret, provide the FastAPI base URL and use the embedded PyCaret setup interface.

Tips & Best Practices

1. Impute missing values before training models that do not handle missingness well.
2. Remove highly correlated predictors to reduce redundancy.
3. Use upsampling when working with imbalanced classification outcomes.

6.3 Train Model

1. Open the Train Model Section

Go to **Machine Learning** then **Train Model**.

The training interface differs depending on whether you selected Caret or PyCaret.

2. Configure Training Options

For Caret, choose relevant settings such as:

- evaluation metric
- cluster usage
- ensemble options

You can also customize train-control settings such as:

- resampling method
- number of folds
- repeats
- search strategy
- verbose iteration
- prediction saving
- class probabilities

3. Select the Models to Train

Choose one or more models from the available list.

The platform supports multiple common model families, including:

- linear and logistic models
- random forest
- GBM
- XGBoost
- SVM
- glmnet
- kNN
- neural networks
- PLS
- bagged trees
- GAM

4. Train the Models

Start the training process after selecting the models.

The app fits the models and stores:

- trained model objects
- train performance
- test performance
- model metadata
- session logs

5. Compare Results

Use the comparative results output to evaluate model performance and identify the strongest candidate.

6. Use the PyCaret Training Interface if Applicable

If PyCaret was selected, the app uses the embedded PyCaret training workflow instead.

Tips & Best Practices

1. Start with a small number of models before running a large comparison.
2. Choose evaluation metrics that match your study goal rather than relying only on defaults.

6.4 Validate and Deploy Model

1. Open the Validate and Deploy Model Section

Go to **Machine Learning** then **Validate and Deploy Model**.

This section is used to review trained models and deploy selected models as endpoints.

2. Review the Available Trained Models

The deployment panel displays the trained models associated with the current dataset/session.

3. Select the Model to Deploy

Choose:

- the session
- the metric
- the model
- the specific trained model instance to deploy

4. Provide the Hostname

Enter the hostname or target deployment address where the model API should be started.

5. Deploy the Model

Click the deployment button.

The app starts the model API and registers the deployed model.

6. Review the Deployment Output

The section displays deployment-related outputs and status information.

Where available, you can also access API/Swagger-related actions.

Tips & Best Practices

1. Deploy only models that have already been validated as acceptable.
2. Use clear session names and metrics so you can identify the correct deployment candidate.

6.5 Predict / Classify

1. Open the Predict / Classify Section

Go to **Machine Learning** then **Predict / Classify**.

This section uses the deployed model endpoint to score new data.

2. Select the Deployed Model

Choose the deployed model endpoint from the model selector.

3. Choose the Prediction Input Type

Select one of the available prediction methods:

- **Upload file**
- **Use form**

4. If Using Upload File

Download the template file first if needed.

Then upload a dataset that matches the expected prediction columns.

The app checks whether the uploaded file has the required structure before allowing prediction.

5. If Using Form Entry

Choose the form option and enter new records manually.

The app allows you to:

- add one record
- save it
- enter another
- finish the entry process

6. Run Prediction

Click the prediction button.

The app sends the new data to the deployed model endpoint and receives the predicted output.

7. Review the Results

The app displays:

- a predictions table
- a prediction plot

These outputs can be downloaded for reporting or reuse.

Tips & Best Practices

1. Use the template file for batch scoring.
2. Use the form option for quick demonstrations or single-record scoring.
3. Make sure variable names and types match the model input schema exactly.

7. Deep Learning (Transformers)

1. Open the Deep Learning Section

Go to **Deep Learning**.

This section supports API-based deep learning workflows for multiple task types.

2. Choose the Task

Select the task you want to run. Available options include:

- **Object Detection**
- **ASR**
- **Image Classification**
- **Image Segmentation**

The task-specific configuration panel then appears.

3. Select a Processed Dataset

Choose a dataset from the task-specific dataset dropdown.

Datasets in this list come from the backend data-processing pipeline.

4. Select the Model Architecture and Checkpoint

Choose:

- the model architecture
- the pretrained checkpoint

The checkpoint list updates dynamically based on the architecture selected.

5. Enter Run Metadata

Provide:

- **Run Name**
- **Version**

These help identify each training job.

6. Configure Task-specific Settings

The app exposes detailed configuration settings depending on the selected task.

Examples include:

Object Detection

- architecture-specific parameters
- learning rate
- optimizer
- warmup
- momentum
- weight decay
- epochs
- batch size
- image size
- gradient checkpointing
- fp16
- early stopping
- hub/W&B logging

ASR

- pre-split vs custom split
- speaker disjointness
- language and language code
- transcript column

- target sampling rate
- duration limits
- transcript length limits
- outlier filtering
- scheduler and optimizer
- gradient checkpointing
- early stopping
- hub/W&B logging

Image Classification

- pre-split option
- train/dev ratio
- learning rate
- batch size
- image size
- gradient accumulation
- gradient checkpointing
- fp16
- early stopping
- hub/W&B logging

Image Segmentation

- segmentation dataset and architecture choices
- split controls
- training controls
- execution controls
- logging and run settings

7. Start the Training Job

Click the task-specific start button, such as:

- **Start Object Detection Job**
- **Start ASR Job**
- **Start Image Classification Job**
- **Start Image Segmentation Job**

The app submits the training job to the backend API.

8. Monitor Job Progress

The app supports live job monitoring through backend polling.

This allows you to follow:

- job status
- current stage
- logs
- progress updates

9. Upload a Deep Learning Dataset

The section also supports dataset management for deep learning tasks.

You can:

- upload a zipped dataset
- assign a dataset name
- assign a task type
- submit it to the backend
- wait for the backend to process it

Once processing is complete, the dataset becomes available in the task-specific dataset dropdown.

Tips & Best Practices

1. Make sure the uploaded dataset matches the selected task type.
2. Start with default parameters before attempting advanced tuning.
3. Use clear run names and versions to keep experiments organized.
4. Only enable advanced hub or W&B options if those services are already configured.

TO DO
Pop-up, tooltip to show more details
Custom visusalizations